

Superconducting **MKID** Detectors at Paris Observatory

Development of a Novel Optical Photon Detector



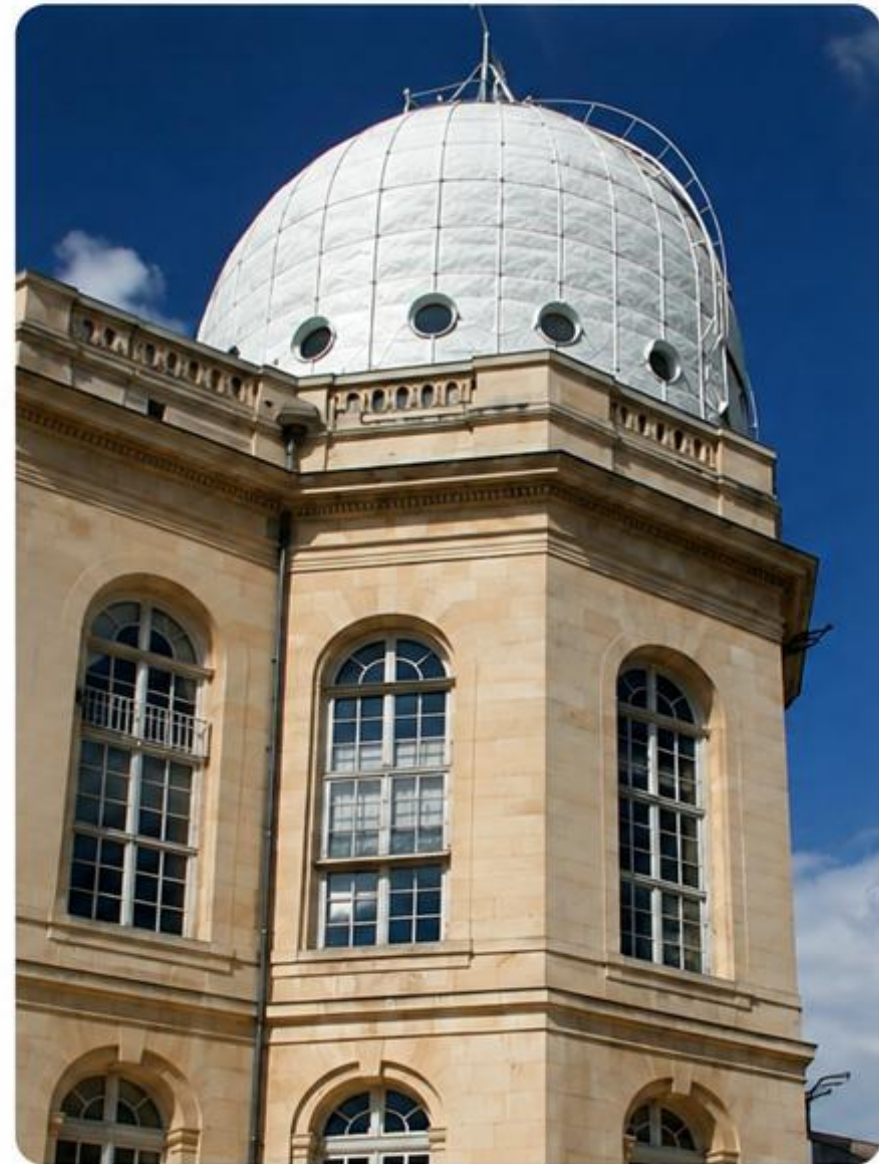
Design & simulation



Cleanroom fabrication



Cryogenic testing



The Challenge:

Seeing Extremely Faint Light

1

**Dwarf galaxies**

Astronomers need to observe ultra-faint dwarf galaxies.

2

**CCD / CMOS**

Existing sensors (CCD / CMOS) are not sensitive enough for these faint signals.

3

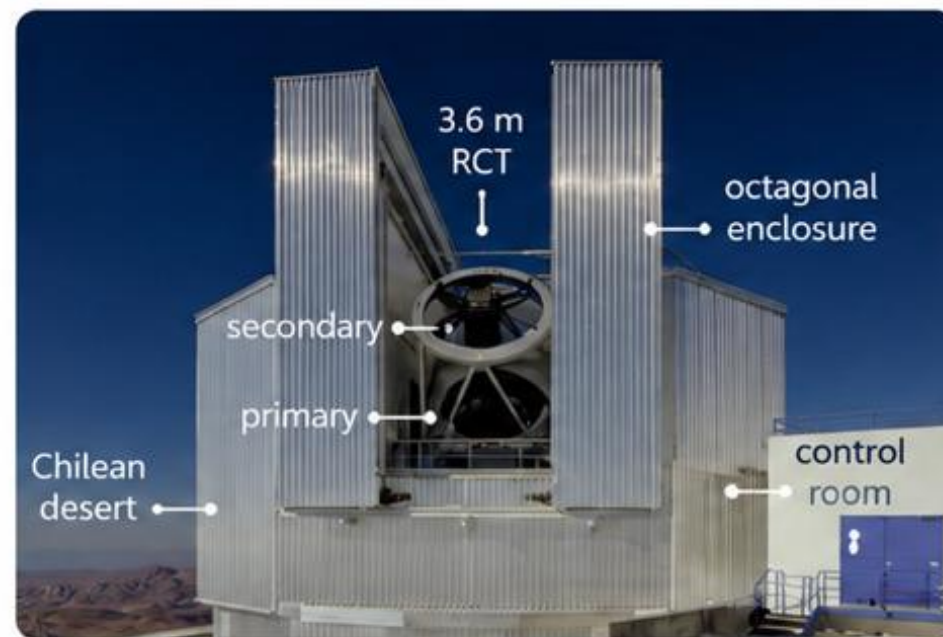
**Count photons**

A detector able to count single photons is required.

4

**MKID technology**

Microwave Kinetic Inductance Detector technology.



NTT: Telescope hosting a 20 000 MKID matrix



Goal: detect individual photons from extremely faint astronomical objects.

What is a MKID?

A MKID (Microwave Kinetic Inductance Detector) is a superconducting sensor that detects **single photons** using a change in **kinetic inductance**.

CCD/CMOS



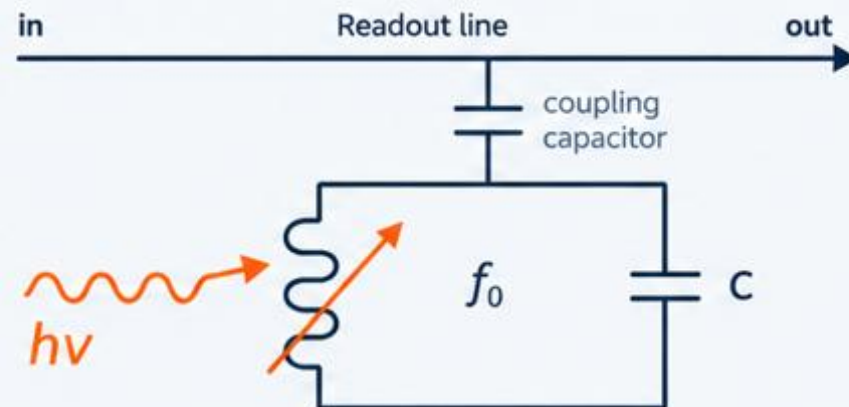
MKID

Same goal:
detect light,
different physics!

Sensor	Operating T	Sensitivity
	0.1 K (very cold)	 Single photon
	~100 K (warmer)	 ~1000× less sensitive

How it works

Ultra-cold
(0.1 K)



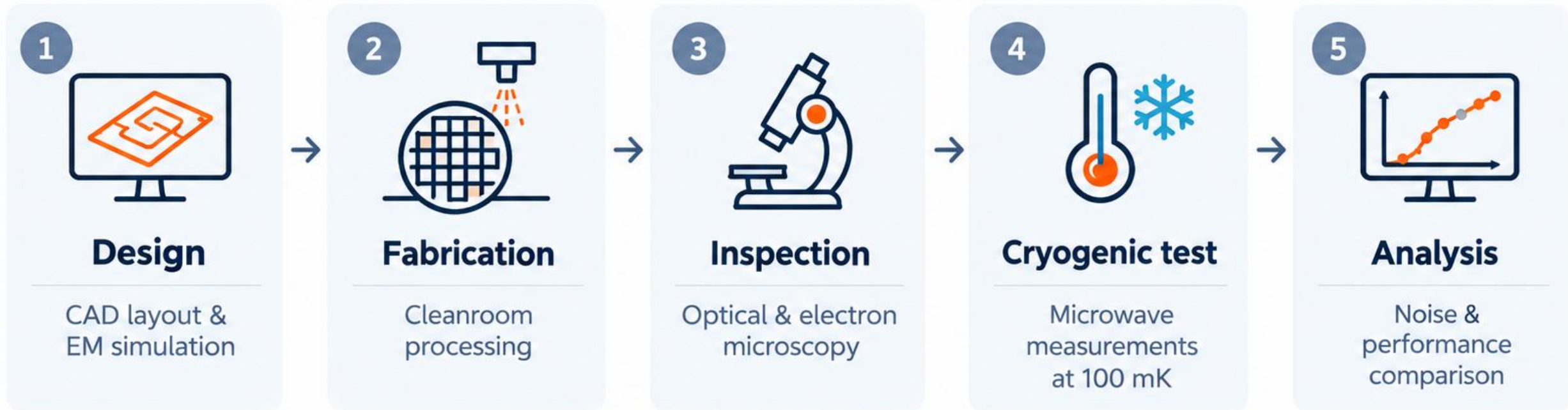
- 1 A photon arrives
- 2 Its **kinetic inductance L_k** changes
- 3 The **resonance frequency f_0** shifts

- 4 We read this shift
→ **photon detected**
(arrival time + energy)



From Design to Test

My role across the detector development cycle



Objective: connect detector geometry to measured performance.

Design & Simulation



AutoCAD

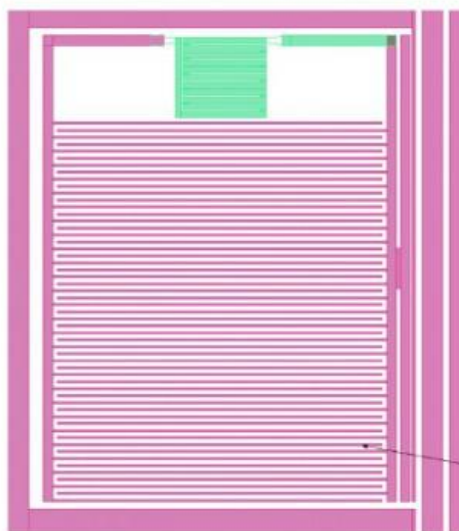
Photomask
geometry



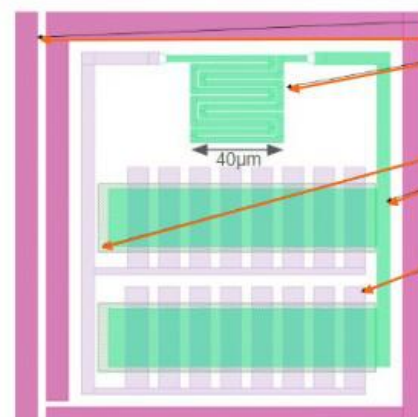
EM simulation

Resonance
frequency &
quality factor

IDC architecture

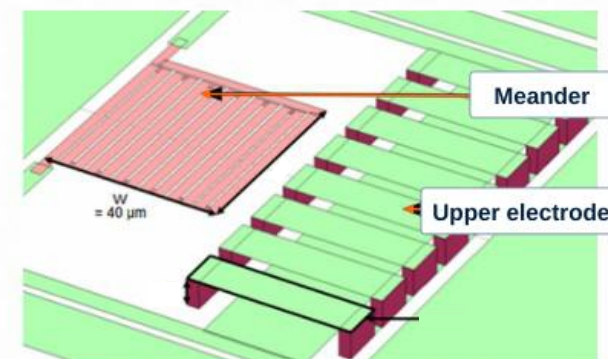


CPV architecture



CPW readout line
Inductive meander
Sacrificial layer
Lower electrode
Upper electrode

CPV 3D CAD view



Suspended upper electrodes above the
lower electrode

Photomask layouts and detector geometry

1

**Design**

2

**Fabrication**

3

**Inspection**

4

**Cryogenic test**

5

**Analysis**

Hands-on cleanroom fabrication

Wafer preparation



Start with a clean wafer

1



Design

2



Fabrication

3



Inspection

4



Cryogenic test

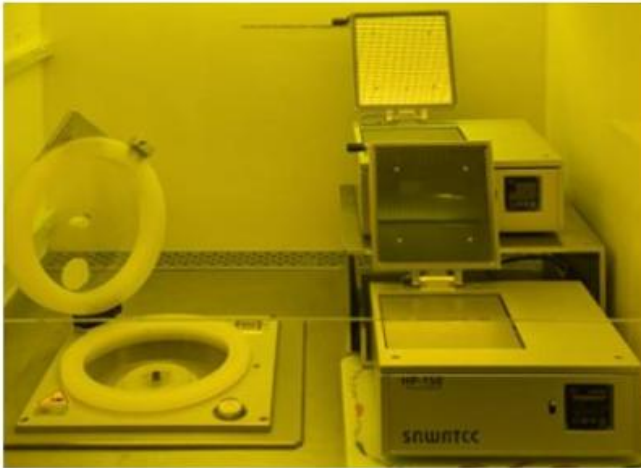
5



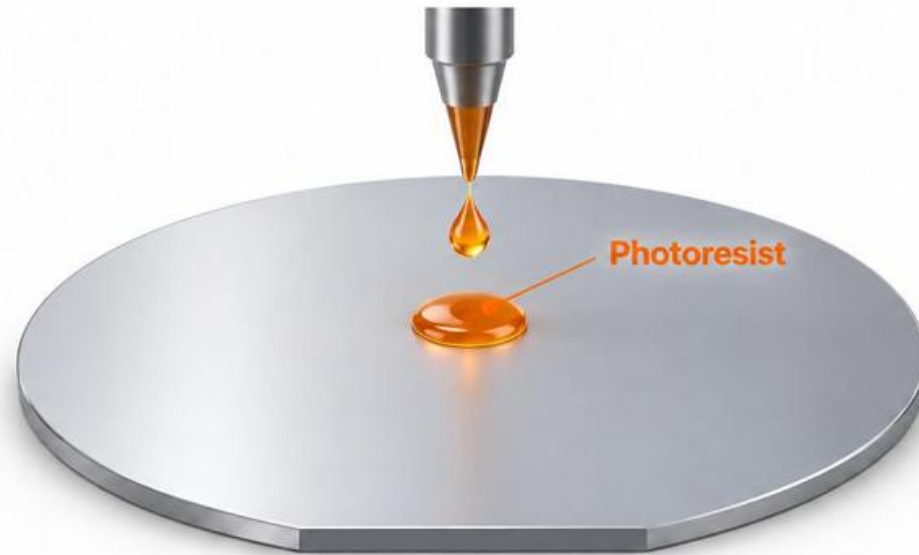
Analysis

Photoresin station

Resin dispensing



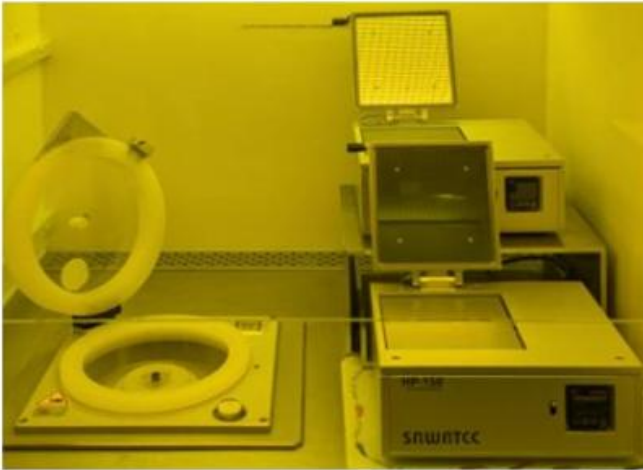
Cleanroom equipment



Coat the wafer with photosensitive resin

Photoresin station

Spin coating



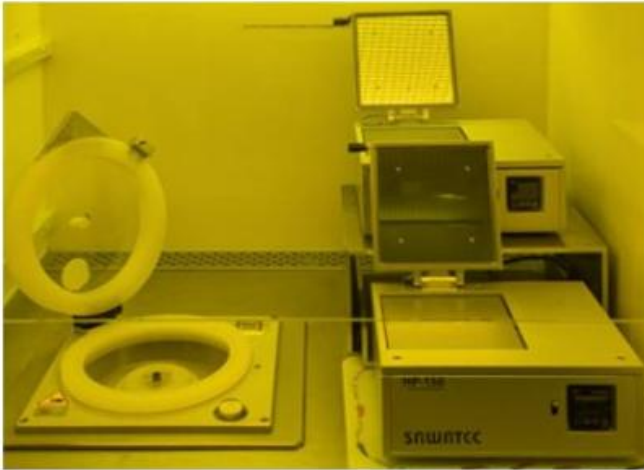
Cleanroom equipment



Spread the resin into a thin, uniform film

Photoresin station

Uniform coating



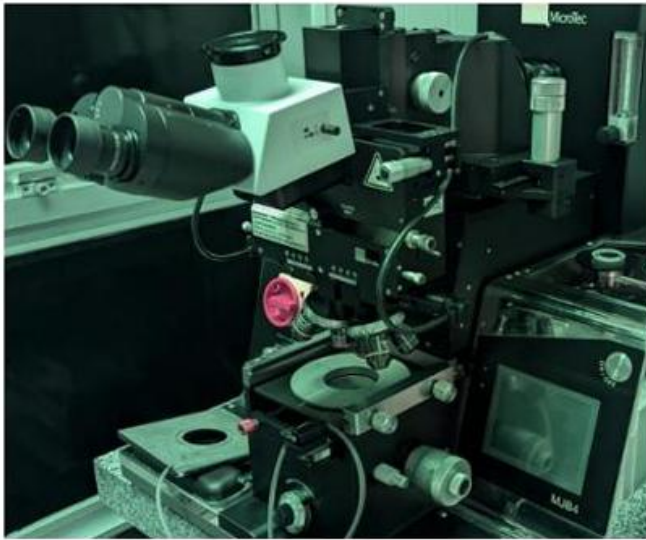
Cleanroom equipment



A uniform photosensitive film covers the wafer

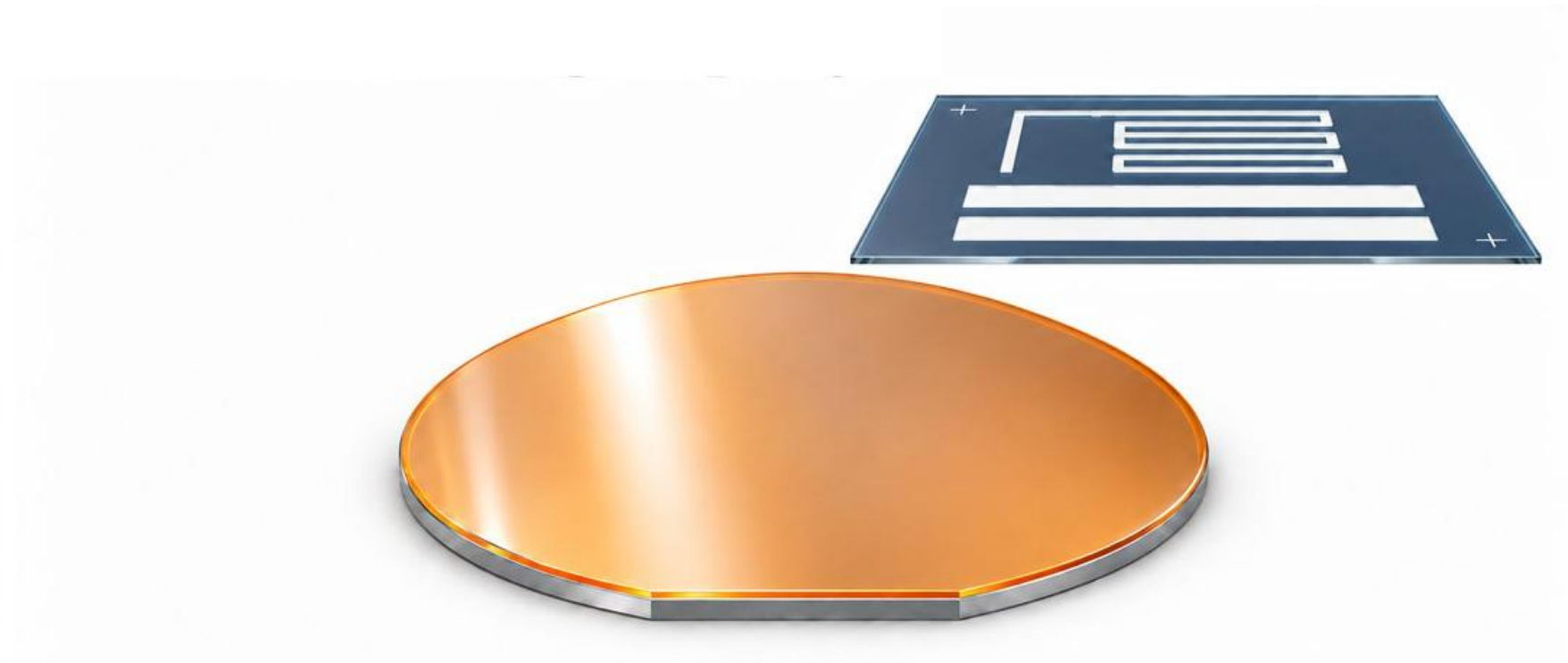
Mask aligner

Photomask introduction



Cleanroom equipment

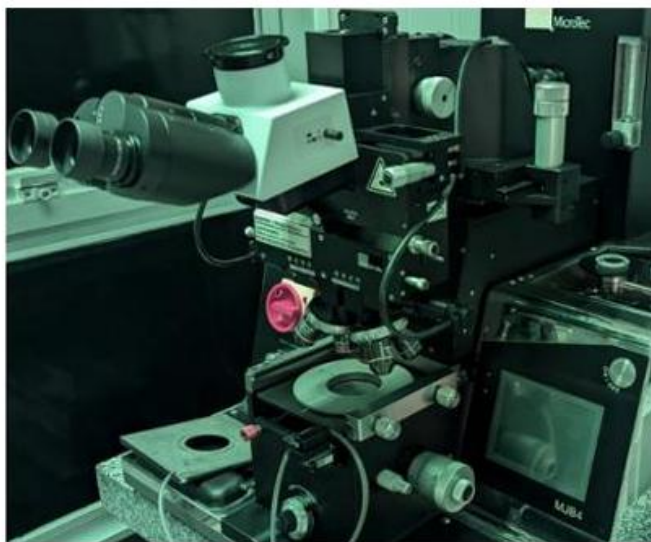
Simplified single-layer example



A photomask carries the simplified CPV pattern

Mask aligner

Mask positioning



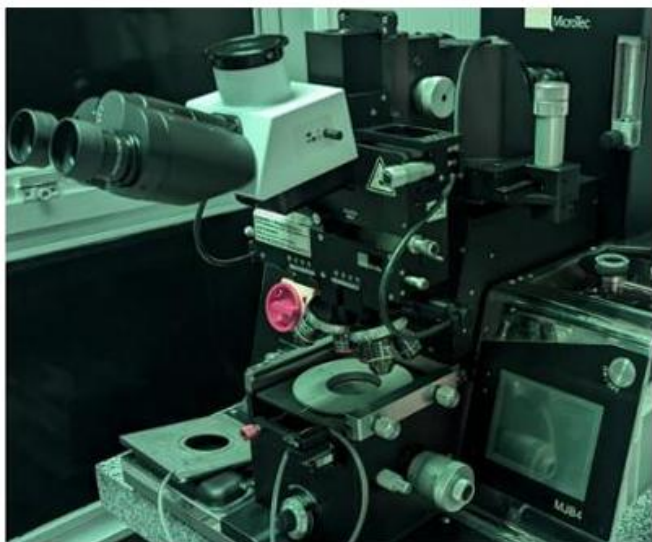
Cleanroom equipment



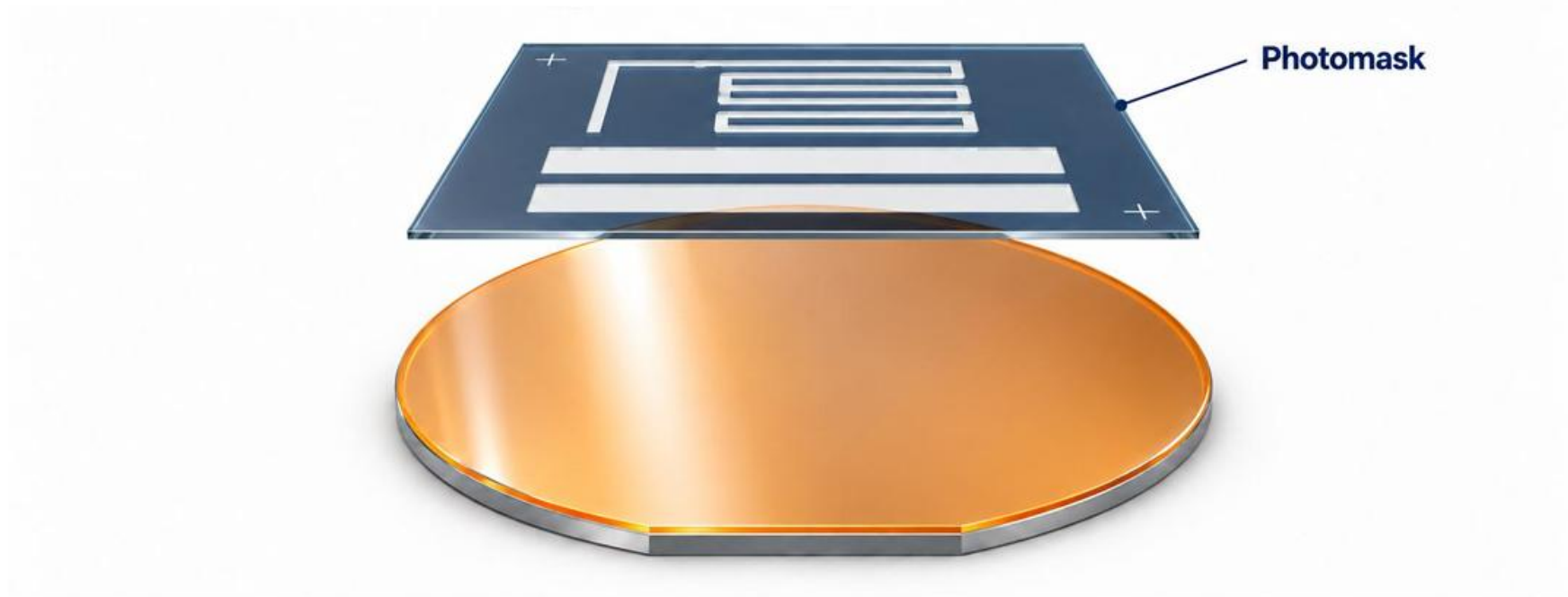
Position the mask above the wafer

Mask aligner

Mask alignment



Cleanroom equipment



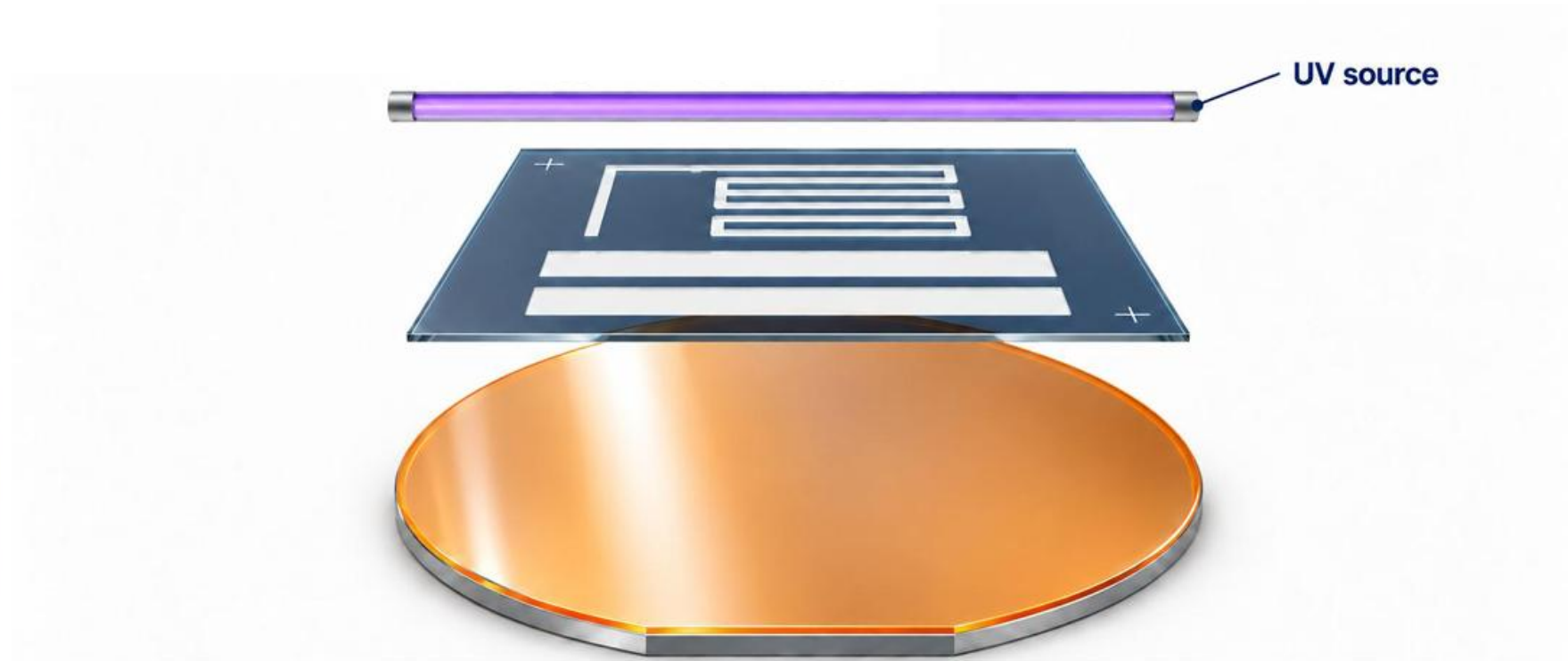
Align the mask above the coated wafer

Mask aligner

UV source



Cleanroom equipment



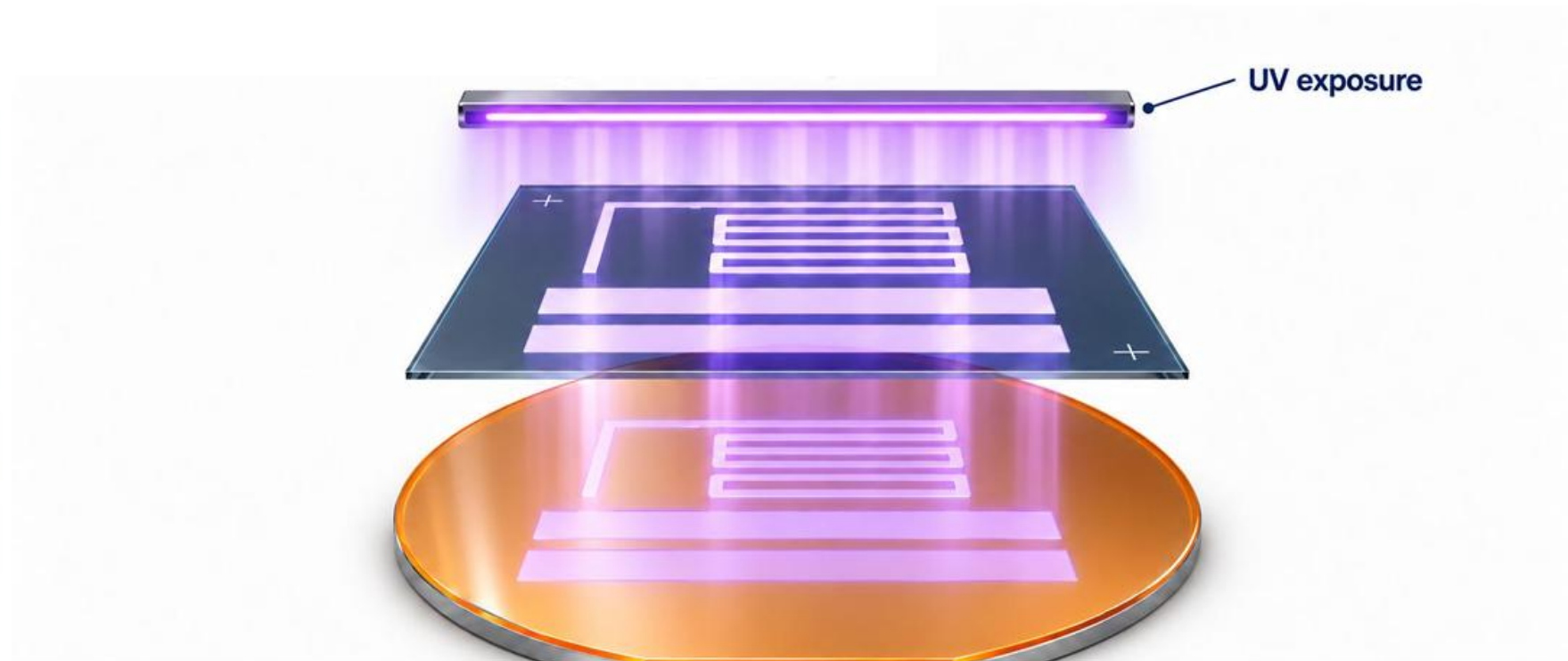
Prepare the UV exposure

Mask aligner

UV exposure



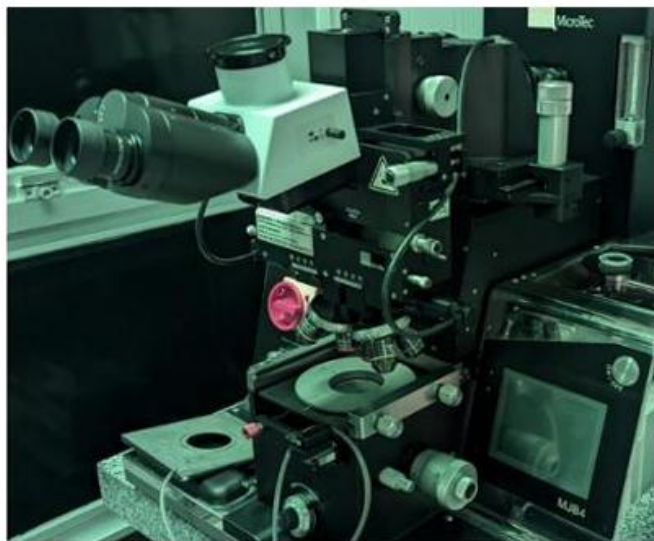
Cleanroom equipment



UV light exposes the resin through the mask openings

Mask aligner

Latent image



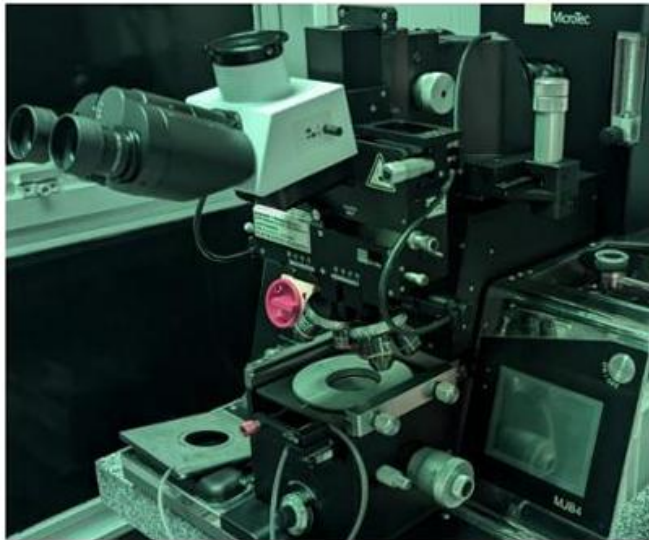
Cleanroom equipment



Exposure creates a latent image in the resin

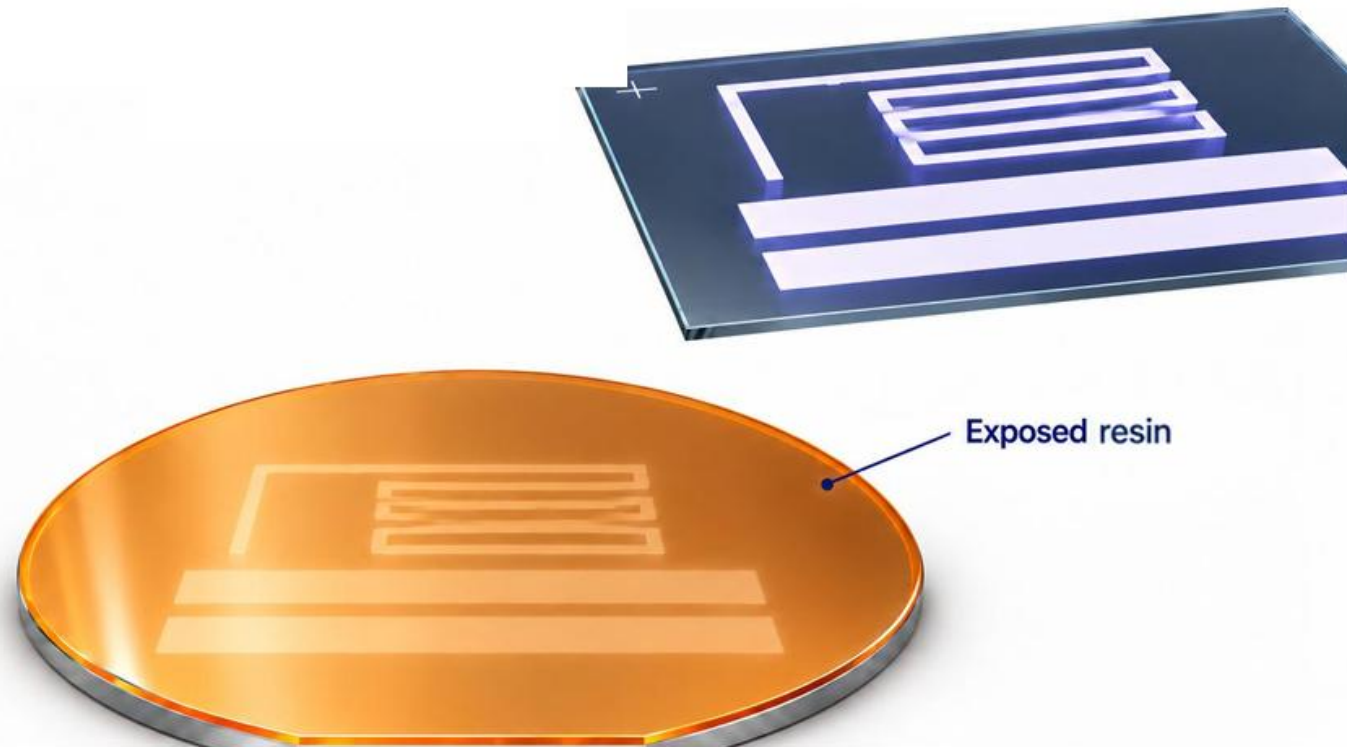
Mask aligner

Mask removal



Cleanroom equipment

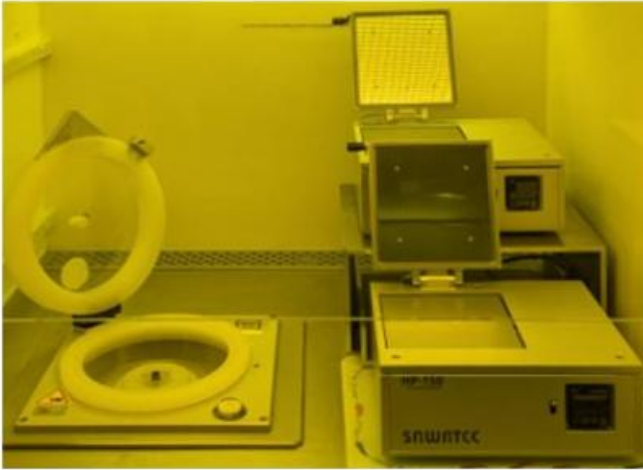
Simplified single-layer example



Remove the mask before development

Photoresin station

Development



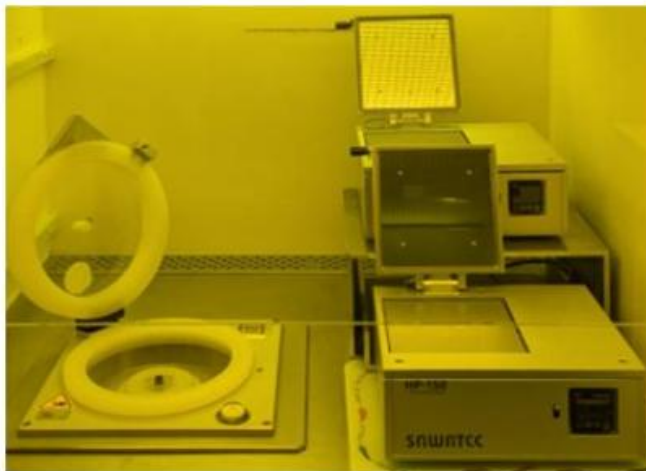
Cleanroom equipment



Development removes the exposed positive resin

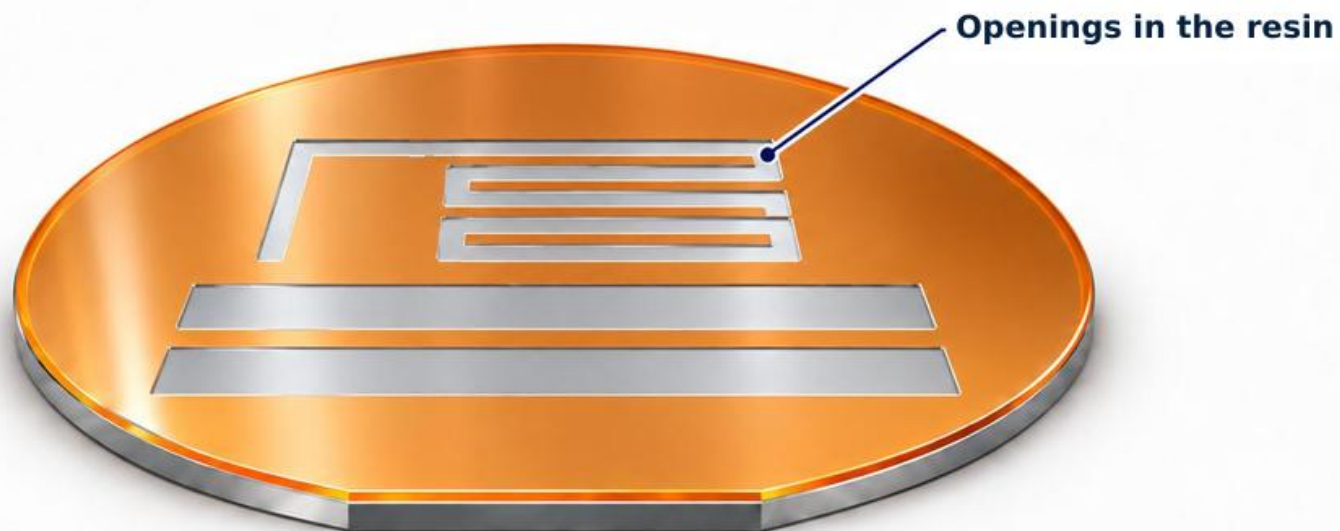
Photoresin station

Patterned resin



Cleanroom equipment

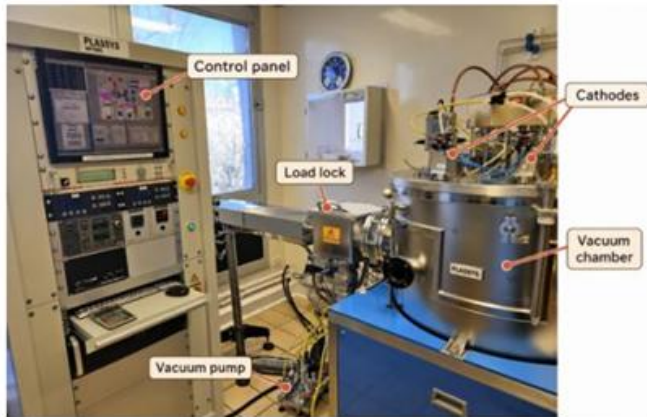
Simplified single-layer example



The remaining resin forms a stencil for deposition

Sputtering system

Vacuum loading



Cleanroom equipment

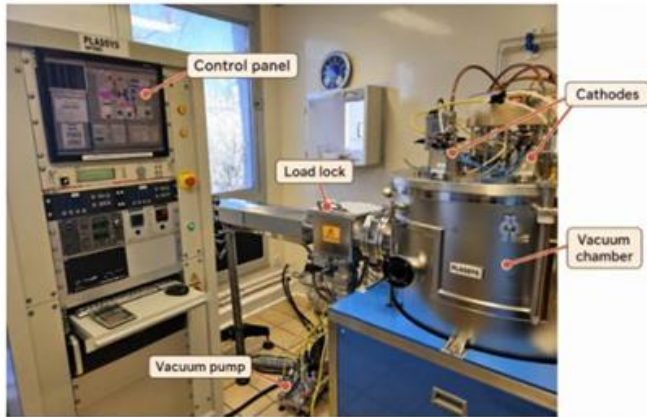
Simplified single-layer example



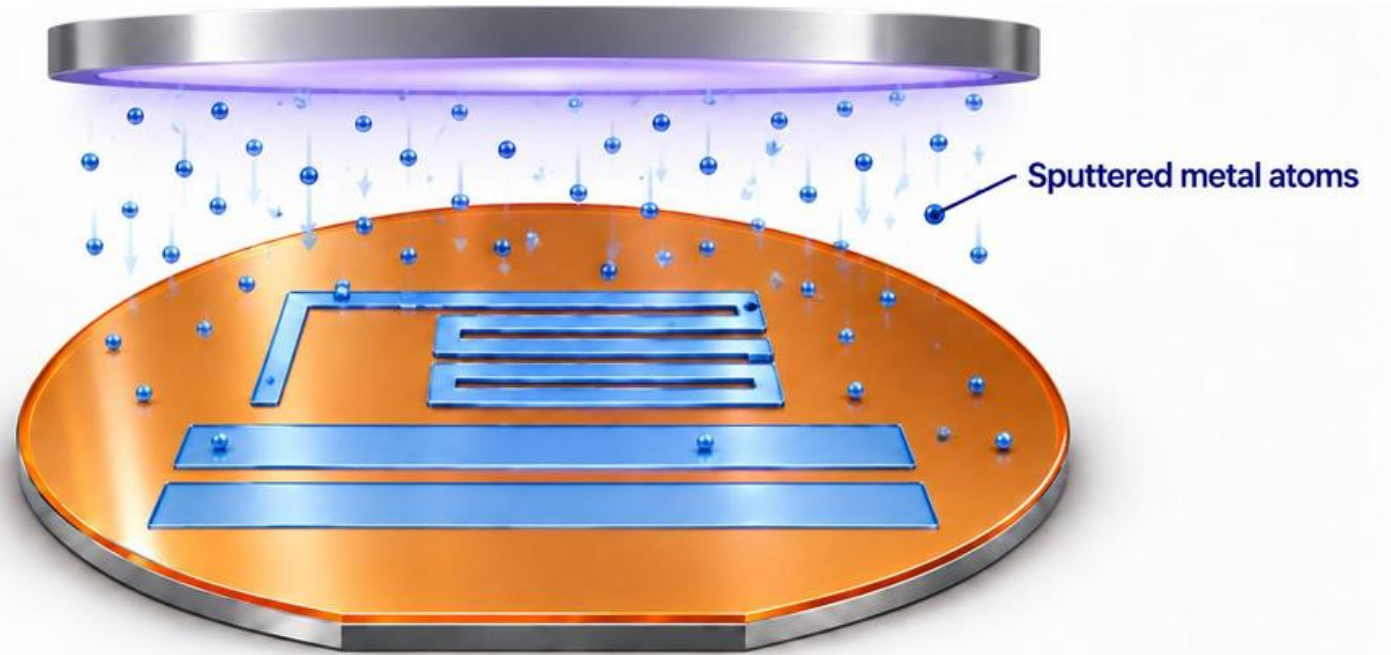
Place the patterned wafer in the vacuum chamber

Sputtering system

Metal deposition



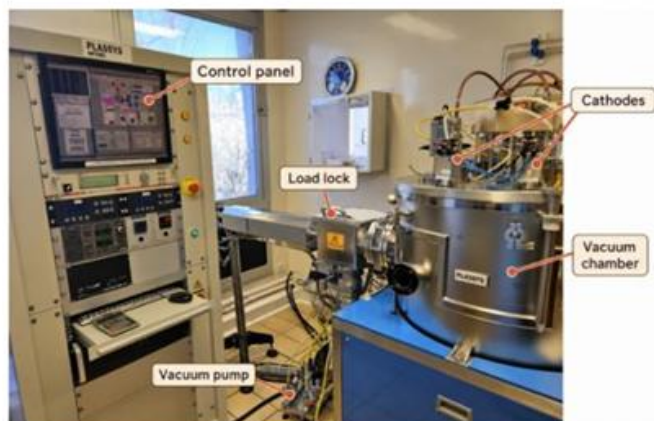
Cleanroom equipment



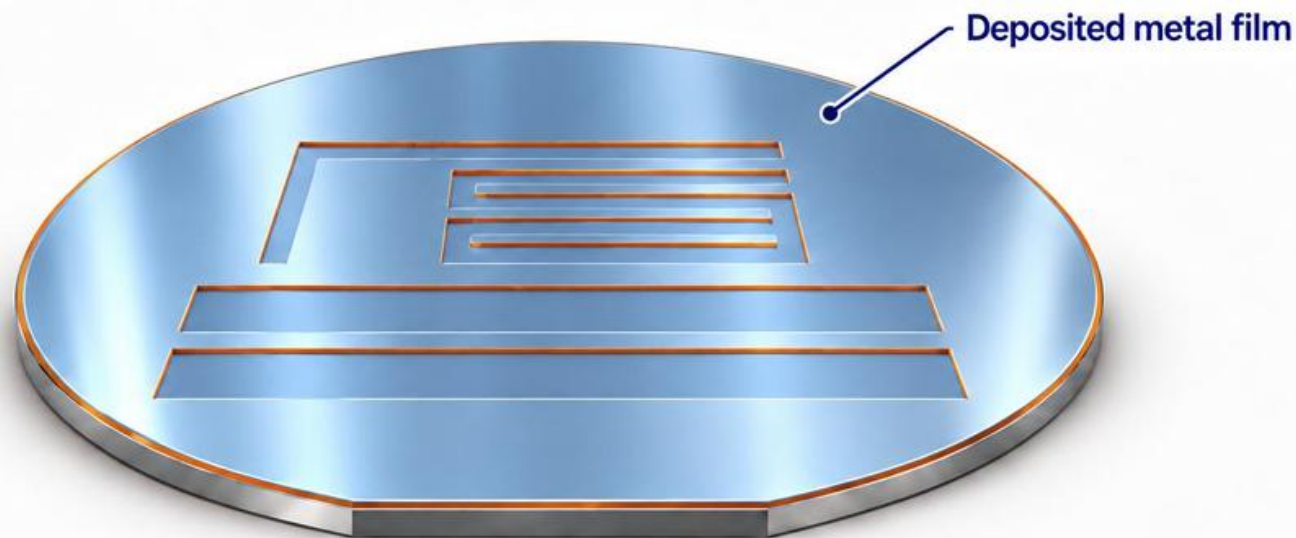
Sputtered atoms coat the resin and the exposed wafer

Sputtering system

Deposited film



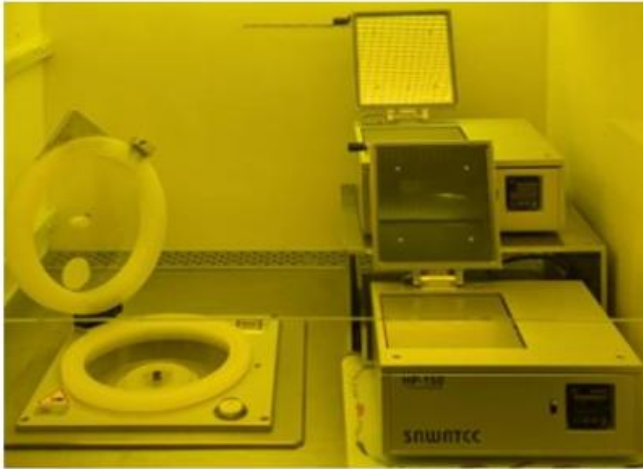
Cleanroom equipment



Metal also covers the top of the resin

Photoresin station

Lift-off



Cleanroom equipment

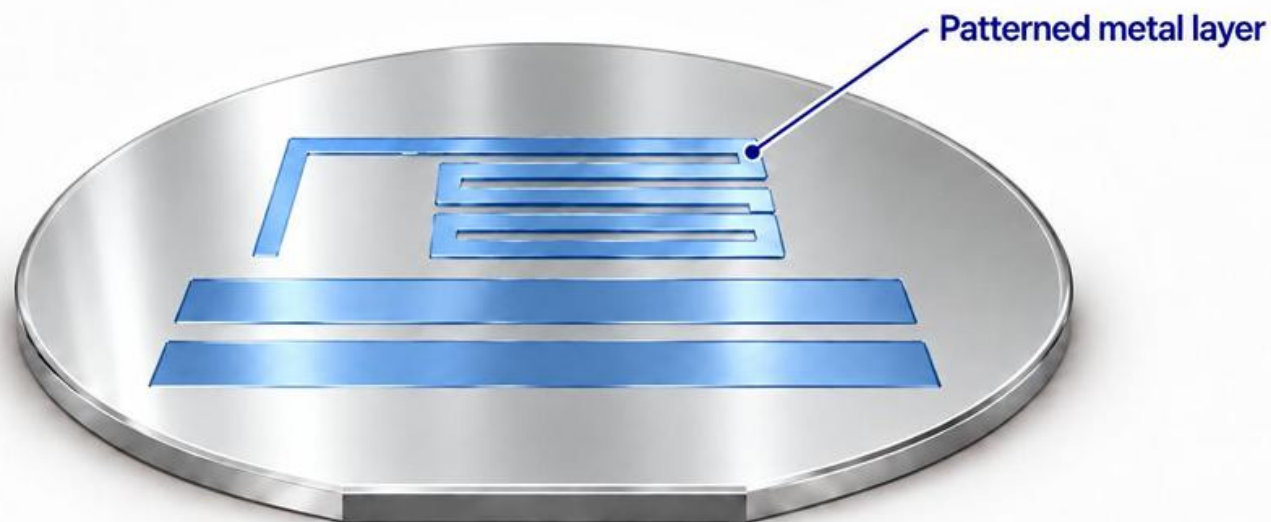
Simplified single-layer example



Lift-off: dissolve the resin and remove the metal above it

Hands-on cleanroom fabrication

Patterned metal layer



Only the patterned metal remains on the wafer

1



Design

2



Fabrication

3



Inspection

4



Cryogenic test

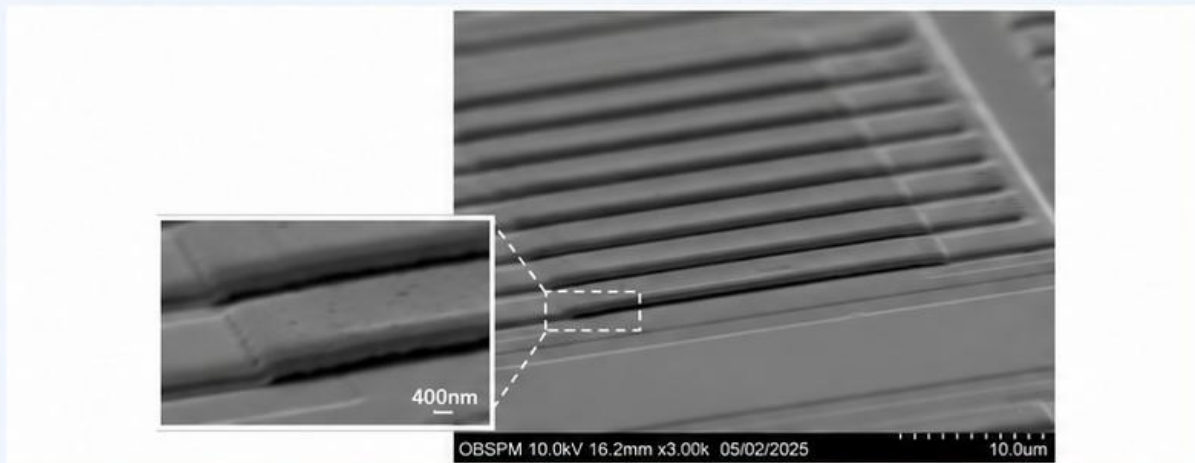
5



Analysis

Inspect, Cool & Measure

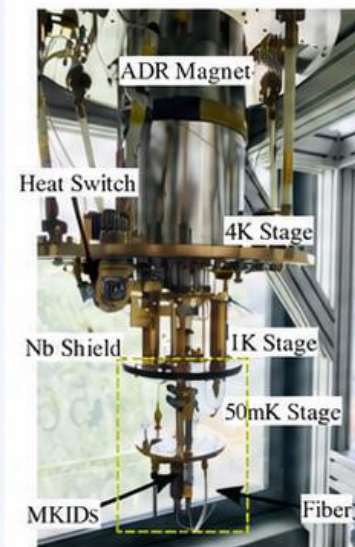
Electron microscopy



Electrode gap inspection

Check the fabricated geometry before testing.

Cryogenic verification



100 mK

Detector operating temperature



VNA sweep

S21 dip at resonance (f_{res}).

1



Design



2



Fabrication



3



Inspection



4



Cryogenic test

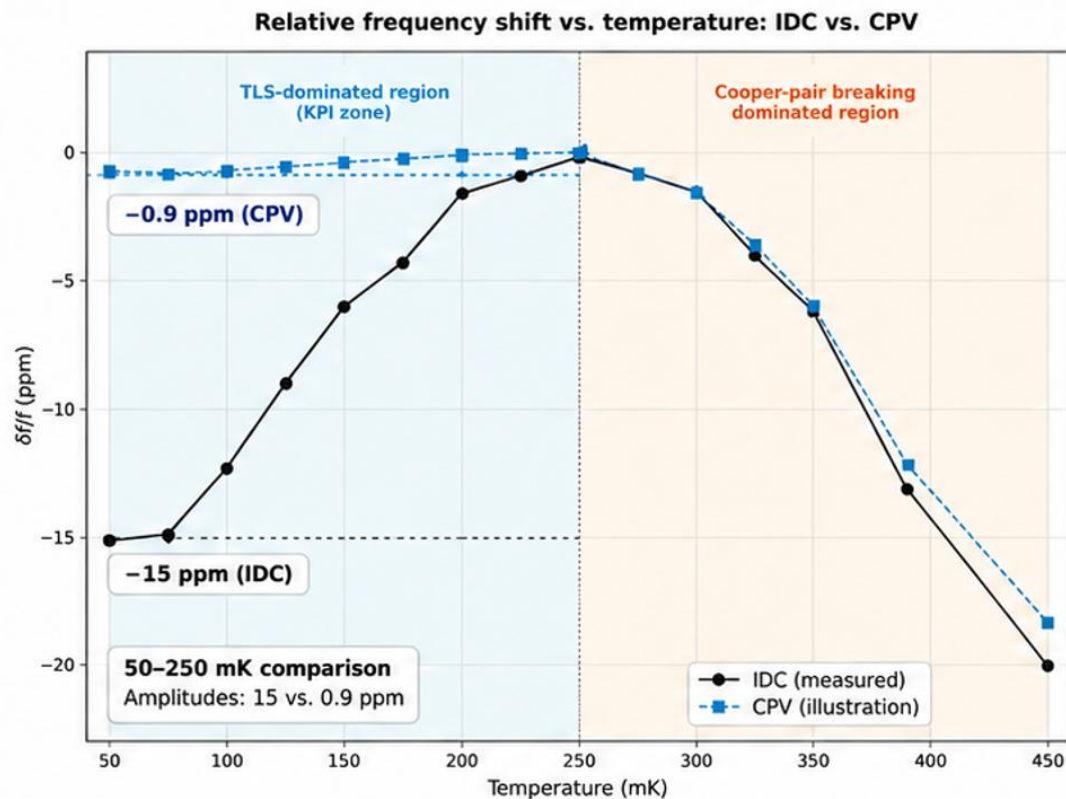


5



Analysis

Results: A Measurable Improvement



CPV trace in the supplied figure is illustrative.

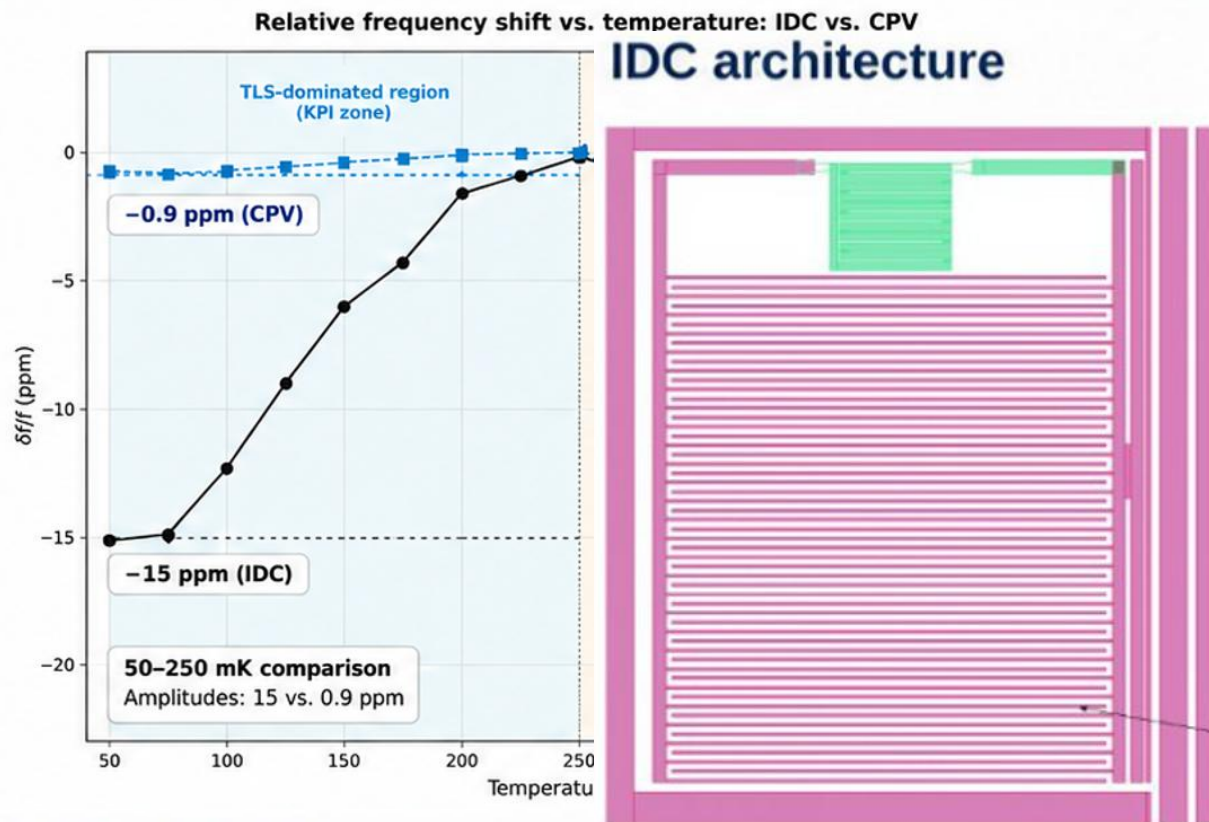
94% ↓
reduction in
TLS-attributed noise

~35%
smaller detector
footprint

Reported IDC–CPV comparison



Results: A Measurable Improvement



94% ↓

reduction in
TLS-attributed noise

~35%

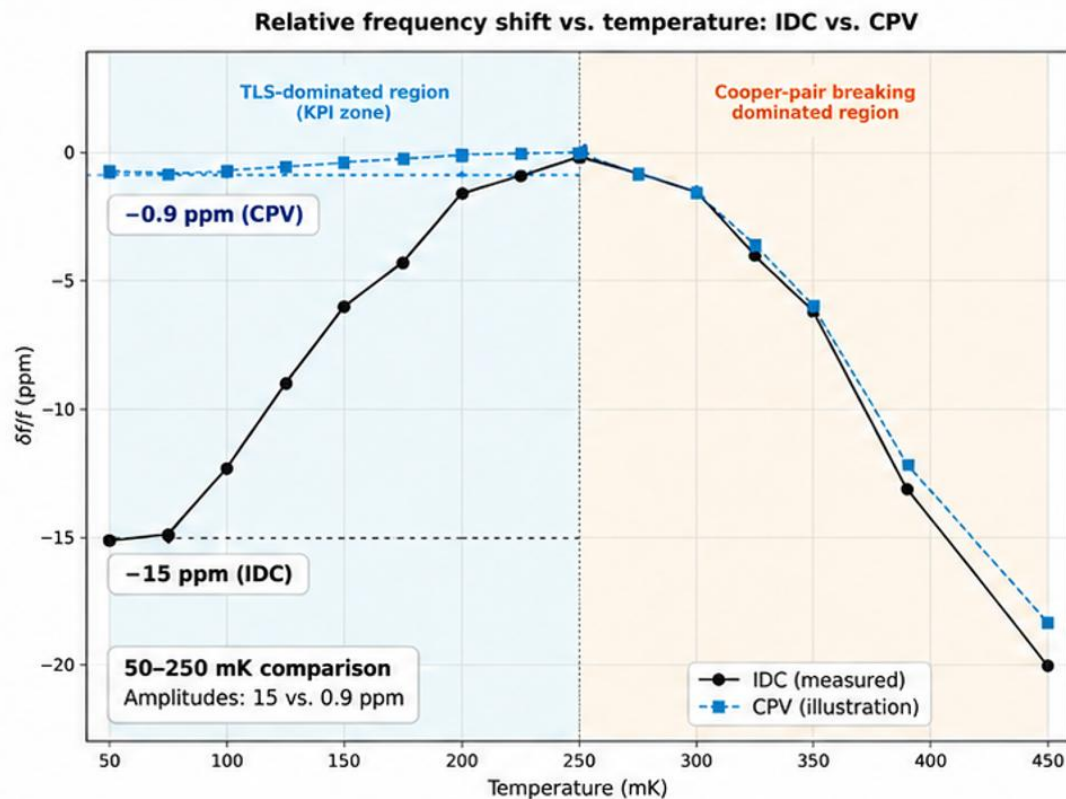
smaller detector
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Reported IDC–CPV comparison

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Results: A Measurable Improvement



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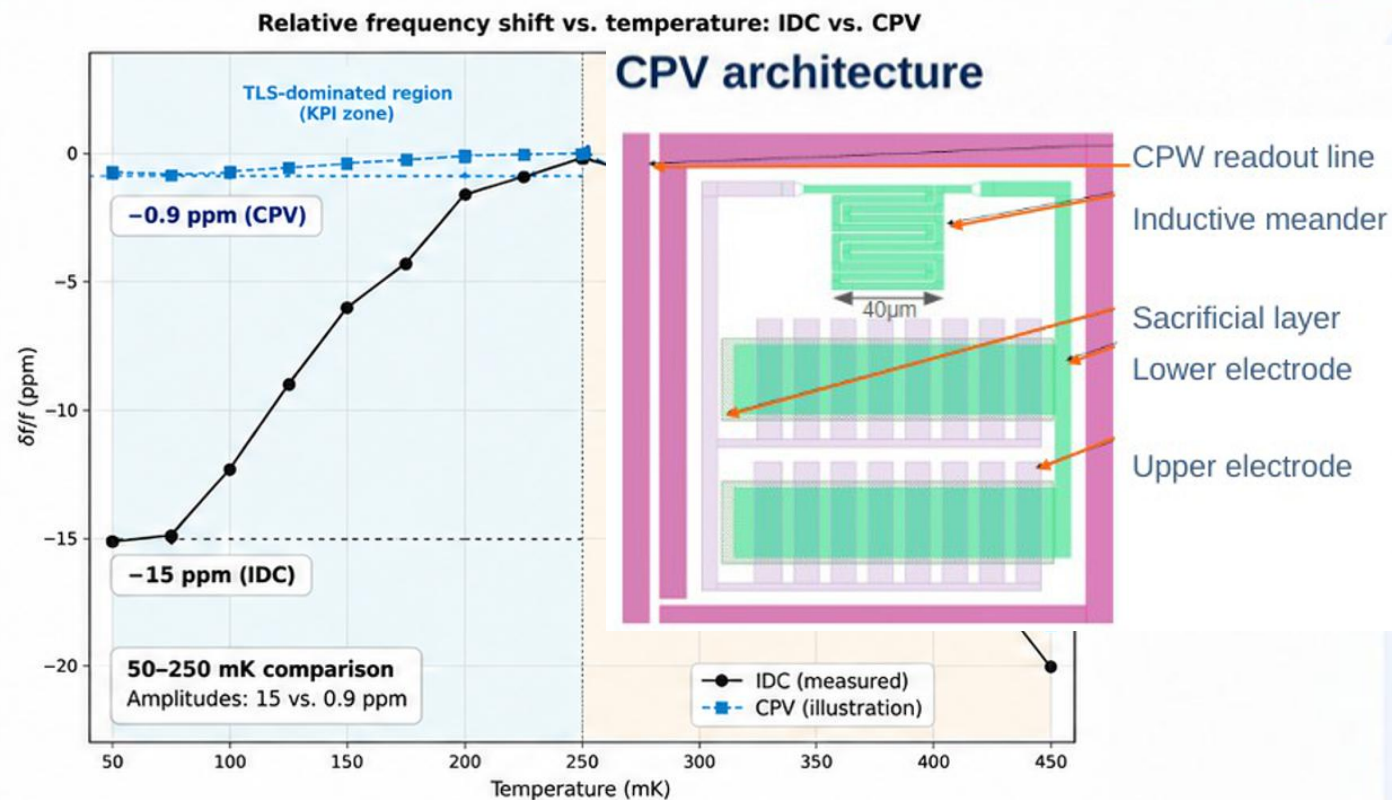
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Results: A Measurable Improvement



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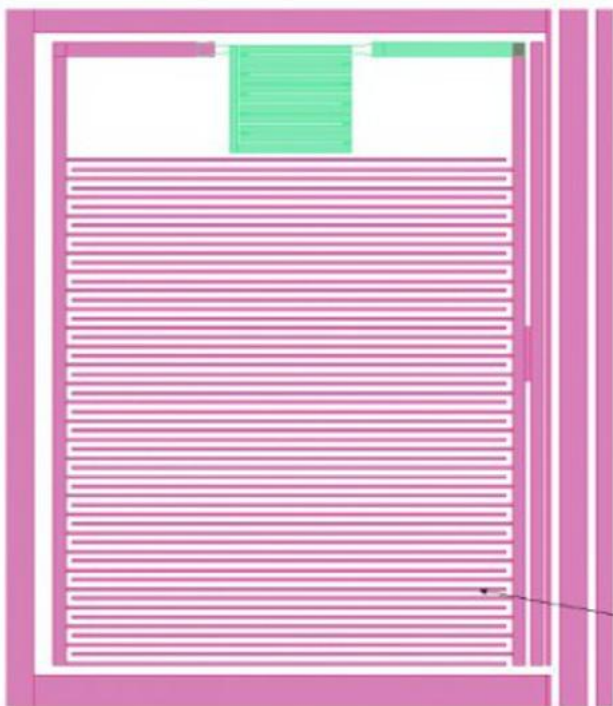
CPV trace in the supplied figure is illustrative.

Reported IDC–CPV comparison

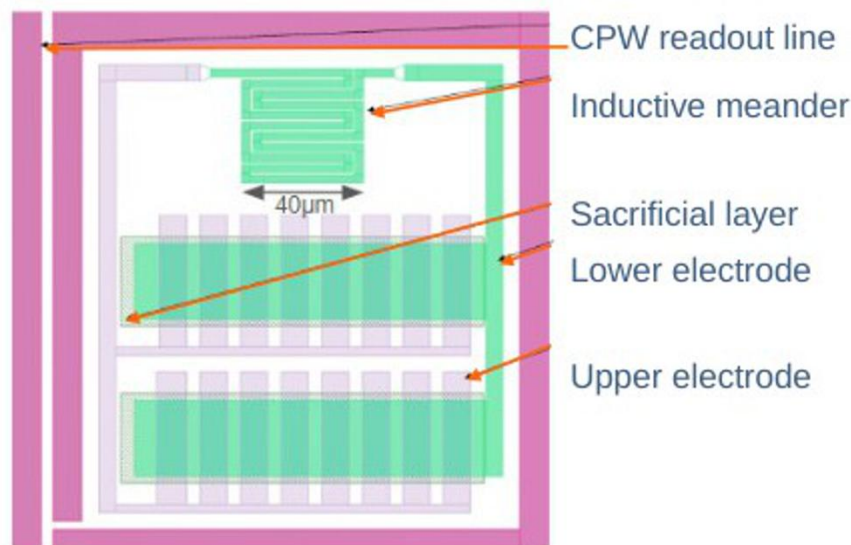


Results: A Measurable Improvement

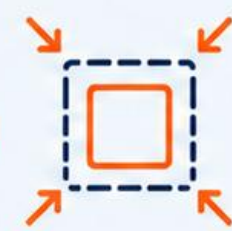
IDC architecture



CPV architecture



94% ↓ 
**reduction in
TLS-attributed noise**

~35% 
**smaller detector
footprint**

Reported IDC-CPV comparison

1



Design



2



Fabrication



3



Inspection



4



Cryogenic test

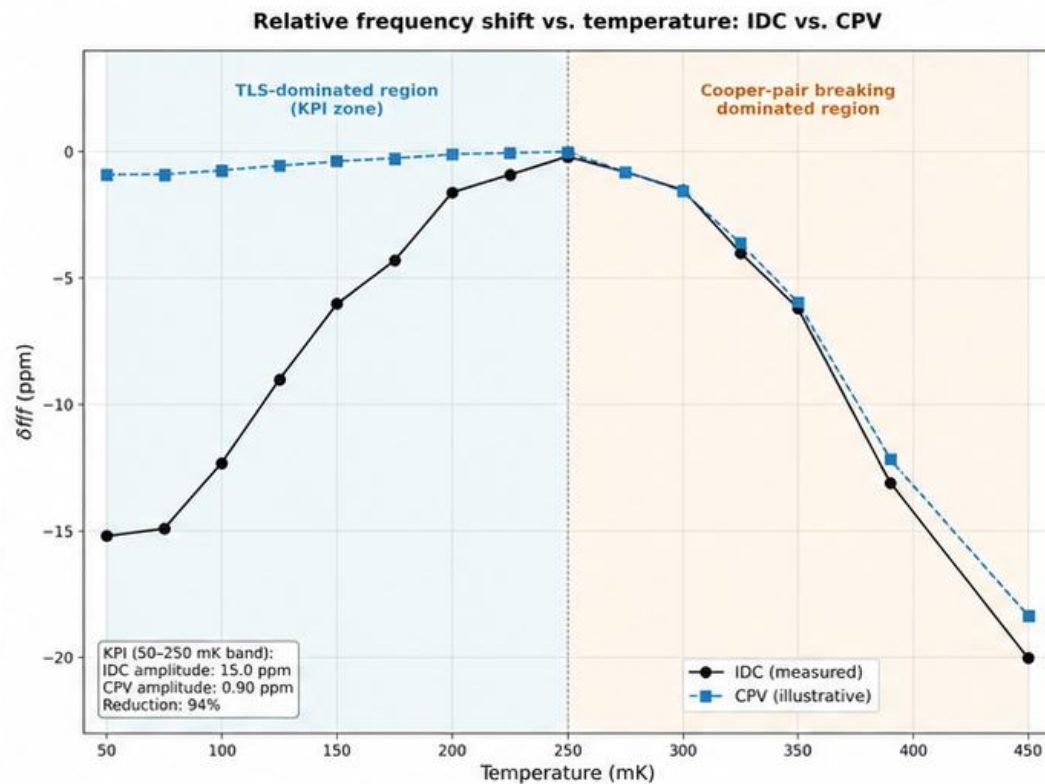


5



Analysis

Results: A Measurable Improvement



94% ↓
reduction in
TLS-attributed noise

~35% ↗
smaller detector
footprint

CPV trace in the supplied figure is illustrative.

Reported IDC-CPV comparison

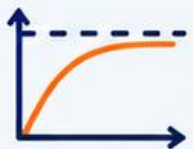


Broader Optical Payload Experience



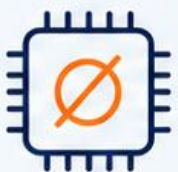
SNR analysis

Model signal-to-noise from detector parameters.



Quantum efficiency

Measure the detected fraction of incident light.

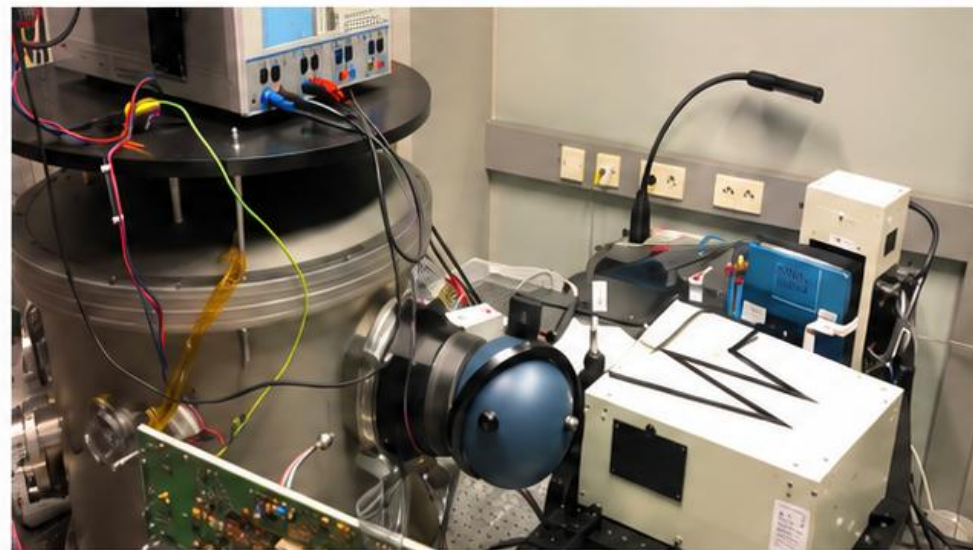


Dark current

Characterize detector response without illumination.



Goal: Experience with optical benches, acquisition systems and vacuum equipment.



Optical test bench

Experience **Relevant** to AIT

Hands-on experience



Cleanroom processing & handling



Detector development



VNA & microwave measurements



Cryogenic & vacuum work



Noise analysis & troubleshooting



Application to AIT



Hardware integration discipline



Optical payload understanding



Electrical and functional testing



Environmental test preparation



Fault finding and test interpretation



From detector design to verified performance.



Questions ?

Appendices

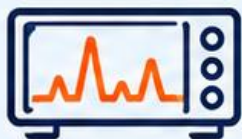
GNSS Operations & Galileo Campaign

Responsible for Signal Observatory operations

Operations & instrumentation



High-gain parabolic antenna
Digitizer



RF measurements
VNA & spectrum analyzer



Python automation
Measurement workflows

Galileo measurement campaign

30 / 30

Targeted satellites
successfully measured



Campaign led in parallel with a
European partner



KPIs derived for cross-comparison



My contribution: led the campaign, operated the observatory and derived comparison KPIs.

Automated SDR Power Calibration

Software-defined radio (SDR) boards — CNES laboratory

Calibration bench



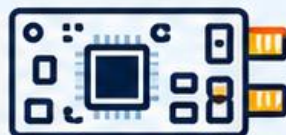
Signal generator



Power sensor



Oscilloscope



SDR boards



GNU Radio Companion

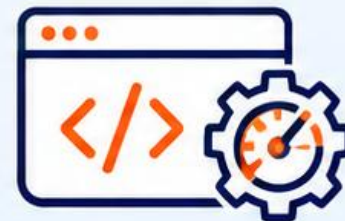


Python

My main achievement

Python automation

Developed a tool that automatically power-calibrates the SDR boards available in the CNES laboratory.



From instrument control to automated calibration



Hands-on experience in RF instrumentation, SDR systems and test automation.